

Start coding or [generate](#) with AI.

ta visualiazation means to seen the data on the screen to representing the user needs

```
!pip install numpy
```

Requirement already satisfied: numpy in /usr/local/lib/python3.12/dist-packages (2.0.2)

```
import numpy as np
print(np.__version__)
```

2.0.2

```
a=np.array([1,2,3,4])
print(a)
```

[1 2 3 4]

```
b=np.array([[1,2,3],
            [4,5,6],
            [7,8,9]])
print(b)
```

[[1 2 3]
[4 5 6]
[7 8 9]]

```
c=np.array([
    [
        [1,2,3],
        [4,5,6],
        [7,8,9]
    ],
    [
        [10,11,12],
        [13,14,15],
        [16,17,18]
    ]
])
print(c)
```

[[[1 2 3]
[4 5 6]
[7 8 9]]

[[10 11 12]
[13 14 15]
[16 17 18]]]

```
a=np.ones([2,3])
b=np.zeros([2,3])
c=np.full([2,3],9)
e=np.eye(4)
print(a,b,c,e)
```

[[1. 1. 1.]
[1. 1. 1.]] [[0. 0. 0.]
[0. 0. 0.]] [[9 9 9]
[9 9 9]] [[1. 0. 0. 0.]
[0. 1. 0. 0.]
[0. 0. 1. 0.]
[0. 0. 0. 1.]]

```
f=np.arange(2,10,3)
print(f)
```

```
[2 6]
```

```
np.linspace(2,10,12)
```

```
array([ 2.          ,  2.72727273,  3.45454545,  4.18181818,  4.90909091,
        5.63636364,  6.36363636,  7.09090909,  7.81818182,  8.54545455,
        9.27272727, 10.          ])
```

Double-click (or enter) to edit

```
np.random.randint(1,5,size=(4,4))
```

```
array([[2, 1, 3, 1],
       [4, 2, 3, 3],
       [2, 1, 4, 2],
       [3, 1, 2, 3]])
```

```
np.empty((3,5))
```

```
array([[2.54965387e-315, 0.00000000e+000, 6.58902806e-310,
        6.58902806e-310, 6.58902806e-310],
       [6.58902806e-310, 6.58902806e-310, 6.58902806e-310,
        6.58902806e-310, 6.58902806e-310],
       [6.58902806e-310, 6.58902806e-310, 6.58902806e-310,
        6.58902806e-310, 1.26480805e-321]])
```

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```
arr=np.array([[1,2,3,4,5,6,7,8]])
print(arr.shape)
print(arr.ndim)
print(arr.size)
print(arr.dtype)
print(arr.itemsize)
print(arr.nbytes)
```

```
(1, 8)
2
8
int64
8
64
```

```
arr=np.array([10,20,30,40,50,60,70,80])
print(a[1:4])
print(a[:3])
print(a[::2])
print(a[::-1])
```

```
[[1. 1. 1.]]
[[1. 1. 1.]
 [1. 1. 1.]]
[[1. 1. 1.]]
[[1. 1. 1.]
 [1. 1. 1.]]
```

```
m=np.array([[1,2,3],
            [4,5,6],
            [7,8,9]])
print(m[0,0])
print(m[1,2])
```

```
print(m[0,:])  
print(m[:,1])  
print(m[0:3, 1:3])
```

```
1  
6  
[1 2 3]  
[2 5 8]  
[[2 3]  
 [5 6]  
 [8 9]]
```

```
a=np.array([10,25,8,42,17])  
mask=a>15  
print(mask)  
print(a[mask])  
print(a[a>15])
```

```
[False  True False  True  True]  
[25 42 17]
```